

Shirin Shoushtari

St. Louis, MO • s.shirin@wustl.edu • [Website](#) • [Google Scholar](#) • [LinkedIn](#)

Education	Washington University in St. Louis <i>Ph.D. Candidate, Electrical & Systems Engineering</i> Advisor: Prof. Ulugbek S. Kamilov	St. Louis, MO Jan 2022 – Expected 2026
	Sharif University of Technology <i>M.Sc., Electrical Engineering – Communication Systems</i> Advisors: Prof. Arash Amini, Prof. Hoda Mohammadzadeh	Tehran, Iran Aug 2018 – Feb 2021
	Sharif University of Technology <i>B.Sc., Electrical Engineering</i> Advisor: Prof. Hamid Behroozi	Tehran, Iran Aug 2012 – May 2017

Research Interests	I develop theory and algorithms for deep learning-based imaging inverse problems , with emphasis on optimization-driven methods , generative models , and robustness under distribution shift . My work spans convergence analysis , out-of-distribution detection , and distribution shift measurement , with applications to computational and quantitative MRI . Going forward, I am particularly interested in flow matching , its deep connections to optimization and minimization theory, and how these connections can be exploited to design principled solvers for imaging inverse problems with provable guarantees .	
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Selected Publications	<i>* denotes equal contribution</i>	
	1. S. Shoushtari , Y. Wang, X. Shi, M. S. Asif, U. S. Kamilov. <i>EigenScore: OOD Detection using Covariance in Diffusion Models</i> . ICLR 2026. [link] [code]	
	2. S. Shoushtari* , J. Liu*, E. P. Chandler, M. S. Asif, U. S. Kamilov. <i>Prior Mismatch and Adaptation in PnP-ADMM with a Nonconvex Convergence Analysis</i> . ICML 2024. [link] [code]	
	3. S. Shoushtari , E. P. Chandler, M. S. Asif, U. S. Kamilov. <i>Unsupervised Detection of Distribution Shift in Inverse Problems using Diffusion Models</i> . arXiv 2025. [link] [code]	
	4. W. Gan, S. Shoushtari , Y. Hu, J. Liu, H. An, U. S. Kamilov. <i>Block Coordinate Plug-and-Play Methods for Blind Inverse Problems</i> . NeurIPS 2023. [link] [code]	
	5. S. Shoushtari , J. Liu, Y. Hu, U. S. Kamilov. <i>Deep Model-based Architectures for Inverse Problems under Mismatched Priors</i> . IEEE JSAT 2022. [link]	

Honors & Awards	– SIAM Travel Award, SIAM Conference on Imaging Science (IS26) (2026) – IEEE CAMSAP Student Paper Award Finalist (2023) – Best Paper Award (4th place), ICBME 2021 — Emotion Recognition via Brain Connectivity – National Entrance Exam (Iran): Top 0.01% B.Sc. (2012); Top 0.1% M.Sc. (2018) – Dean’s Honorary Award, Student Union (2014)	
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Presentations & Talks	– “Out-of-Distribution Detection via Uncertainty Quantification,” Asilomar 2025, Pacific Grove, CA. – “PnP Restoration with Domain Adaptation for Accelerated SANS,” Gordon Research Seminar 2024, Newry, ME. – “Diffusion Models for Phase Retrieval in Computational Imaging,” Asilomar 2023, Pacific Grove, CA.
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Teaching Experience	Teaching Assistant – Signals & Systems WashU, Spring 2025 Teaching Assistant – Optimization WashU, 2023 Teaching Assistant – Statistical Learning Sharif University, 2018
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Mentorship & Supervision	– Xiao Shi (M.S. WashU, 2025 → Ph.D. ECE, UW-Madison) — co-authored EigenScore, ICLR 2026. – Kevin Wang (B.S. WashU) — co-authored EigenScore, ICLR 2026. – Weining Wang (M.S. student, WashU) — co-authored CAMSAP 2025. – Mingi Kang (visiting undergrad, Bowdoin College) — research on quantitative MRI.
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Professional Service	Journal Reviewer: IEEE TSP, IEEE TCI, IEEE TMI Conference Reviewer: NeurIPS, ICLR, CVPR, ICASSP, AISTATS, AAAI, Asilomar, ISCS
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Conference Papers	S. Shoushtari, Y. Wang, X. Shi, M. S. Asif, U. S. Kamilov. <i>EigenScore: OOD Detection using Covariance in Diffusion Models</i>. ICLR 2026. [link] [code] C. Park, S. Shoushtari, H. An, U. S. Kamilov. <i>Measurement Score-Based Diffusion Model</i>. ICLR 2026. [link] [code] S. Shoushtari*, J. Liu*, E. P. Chandler, M. S. Asif, U. S. Kamilov. <i>Prior Mismatch and Adaptation in PnP-ADMM with a Nonconvex Convergence Analysis</i>. ICML 2024. [link] [code] S. Shoushtari, E. P. Chandler, U. S. Kamilov. <i>Moments Matter: Posterior Recovery in Poisson Denoising via Log-Networks</i>. ICASSP 2026. [link] W. Gan, S. Shoushtari, Y. Hu, J. Liu, H. An, U. S. Kamilov. <i>Block Coordinate Plug-and-Play Methods for Blind Inverse Problems</i>. NeurIPS 2023. [link] [code] Z. Zou*, J. Liu*, S. Shoushtari, Y. Wang, W. Gan, U. S. Kamilov. <i>FLAIR: A Conditional Diffusion Framework with Applications to Face Video Restoration</i>. WACV 2025. [link] [code] C. Park*, S. Shoushtari*, W. Gan, U. S. Kamilov. <i>Convergence of Nonconvex PnP-ADMM with MMSE Denoisers</i>. CAMSAP 2023. <i>Best Paper Award Finalist</i>. E. P. Chandler, S. Shoushtari, J. Liu, M. S. Asif, U. S. Kamilov. <i>Overcoming Distribution Shifts in Plug-and-Play Methods with Test-Time Training</i>. CAMSAP 2023. [link] J. Hu, S. Shoushtari, Z. Zou, J. Liu, Z. Sun, U. S. Kamilov. <i>Robustness of Deep Equilibrium Architectures to Changes in the Measurement Model</i>. ICASSP 2023. [link]
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- S. Shoushtari**, J. Liu, U. S. Kamilov. *DOLPH: Diffusion Models for Phase Retrieval*. Asilomar 2023. [\[link\]](#)
- J. Liu, X. Xu, W. Gan, **S. Shoushtari**, U. S. Kamilov. *Online Deep Equilibrium Learning for Regularization by Denoising*. NeurIPS 2022. [\[link\]](#) [\[code\]](#)
- W. Wang, **S. Shoushtari**, E. P. Chandler, M. S. Asif, U. S. Kamilov. *Distribution Shift Estimation in Imaging Inverse Problems*. CAMSAP 2025.
- S. Shoushtari**, Y. Wang, U. S. Kamilov. *Out-of-Distribution Detection via Uncertainty Quantification in Diffusion Models*. Asilomar 2025.
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Journal Papers

- E. P. Chandler, **S. Shoushtari**, B. Wohlberg, U. S. Kamilov. *Closed-Form Approximation of the Total Variation Proximal Operator*. IEEE Transactions on Computational Imaging, 2025. [\[link\]](#)
- S. Shoushtari**, J. Liu, Y. Hu, U. S. Kamilov. *Deep Model-based Architectures for Inverse Problems under Mismatched Priors*. IEEE Journal on Selected Areas in Information Theory, 2022. [\[link\]](#)
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Preprints

- S. Shoushtari**, E. P. Chandler, M. S. Asif, U. S. Kamilov. *Unsupervised Detection of Distribution Shift in Inverse Problems using Diffusion Models*. arXiv 2025. [\[link\]](#) [\[code\]](#)
- E. P. Chandler, **S. Shoushtari**, B. Wohlberg, U. S. Kamilov. *Analysis Plug-and-Play Methods for Imaging Inverse Problems*. arXiv 2025. [\[link\]](#)
- Z. Zou, **S. Shoushtari**, J. Liu, J. Zhang, P. Judge, E. Santana, A. Lim, M. Foston, U. S. Kamilov. *TVCondNet: A Conditional Denoising Neural Network for NMR Spectroscopy*. arXiv 2024. [\[link\]](#)
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References

Prof. Ulugbek S. Kamilov (Ph.D. Advisor)
Leon and Elizabeth Janssen Associate Professor of Electrical and Computer Engineering
University of Wisconsin–Madison
kamilov@wisc.edu • ukmlv.github.io

Prof. M. Salman Asif
Associate Professor, Electrical and Computer Engineering
University of California, Riverside
sasif@ucr.edu • www.ece.ucr.edu/~sasif